

# Welcome to **instats**

*The session will begin shortly.*

# Start

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THIRD MOVEMENT • 35 MINUTES

# Crawling, with manners.

*The two-screen check that decides whether you can crawl a site at all. The five-line Python that does the actual work.*

03

# Read robots.txt before any code runs.

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```
# https://www.csis.org/robots.txt User-agent: * Allow:
/analysis/ Allow: /*?page= Disallow: /admin/ Disallow: /search/
Disallow: /user/ Disallow: /comment/ User-agent: Bytespider
Disallow: / # Verified 2026-04-28. Re-check # before any second-
pass crawl.
```

## THREE THINGS TO FIND

- **User-agent.** Which crawlers each rule block applies to.
- **Allow.** Paths you may fetch.
- **Disallow.** Paths you may not fetch.

## WHY IT MATTERS BEFORE ANY CODE

Robots.txt is a social contract. It does not have legal force in most jurisdictions. It is the first thing a methods reviewer will ask whether you checked. Document the date you read it. Save a copy. Re-check before any second pass.

# The Python is short. The hygiene is the whole job.

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```
# A polite fetch in 8 lines.
import requests, time

UA = 'workshop-demo (mailto:you@univ.edu)'

def polite_fetch(url):
    r = requests.get(url,
        headers={'User-Agent': UA},
        timeout=10)
    if r.status_code != 200:
        return None
    time.sleep(1.0)
    return r.text # raw HTML
```

- **User-Agent.** A real string that identifies you and gives a contact email. Sysadmins should be able to reach you instead of blocking your IP.
- **Status check.** 200 is the only happy path. 403 means blocked. 404 means moved. 429 means slow down. 503 means try later.
- **Timeout.** Default 10 seconds. Without it your script hangs on a slow page and never tells you why.
- **Raw HTML.** Store the bytes, not just the parsed text. Slide 21 explains why.
- **Sleep.** One second between requests is the floor. Slide 22 details politeness limits.

# Crawl one CSIS analysis page together.

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## What we run

Section 9. Define `polite_fetch()`. User-Agent string. Timeout. Status check. Returns raw HTML and metadata.

Section 10. Fetch one article URL. Parse with BeautifulSoup. Pull the headline, publication date, and first paragraphs. Inspect the parsed output. Note the raw HTML length, around 100 KB.

CSIS is server-rendered HTML. No JavaScript execution required. That is why generic Python crawls work here, and why the same approach can fail elsewhere. If your live test fails, the site is telling you to use their public API.

# Save the raw HTML after every fetch.

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## REASON 01

### Parsers change

You will discover next month that the date selector also matches a sidebar element. Re-running the parser on stored HTML is a one-line script. Re-crawling the site is a one-day script and a fresh robots check.

## REASON 02

### Sites change

The article you scraped today may not look the same in six months. Stored raw HTML is the only durable evidence that your data existed in the form you claim.

## REASON 03

### Replication packages

You ship the parser plus the raw HTML. A reviewer running your code in 2030 should not need network access to reproduce your dataset.

*Filename is the URL slug plus a timestamp. Compress as gzip if the corpus grows.*

# Be slower than you think you need to be.

RATES OF FIRE

| PACE      | VERDICT                                |
|-----------|----------------------------------------|
| 0.5 / sec | Polite. Default for production crawls. |
| 1 / sec   | Floor. Today's notebook setting.       |
| 5 / sec   | Rude on most sites. Avoid.             |
| 20 / sec  | Denial of service. Never.              |

THREE IMPLEMENTATION DETAILS

- **Sleep after each request** rather than before the first. The point is to space the load on the server. Spacing your own script is a side effect.
- **Cache responses by URL hash.** If you re-run, do not re-fetch what you already have.
- **Log every request** with a timestamp. If a sysadmin emails you, you can show your traffic pattern from your own logs.

CSIS does not publish explicit rate limits. The default in the demo is one second between requests. Tune slower if your IP starts catching 429 responses. Never speed up.

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FOURTH MOVEMENT • 25 MINUTES

# End-to-end pipeline.

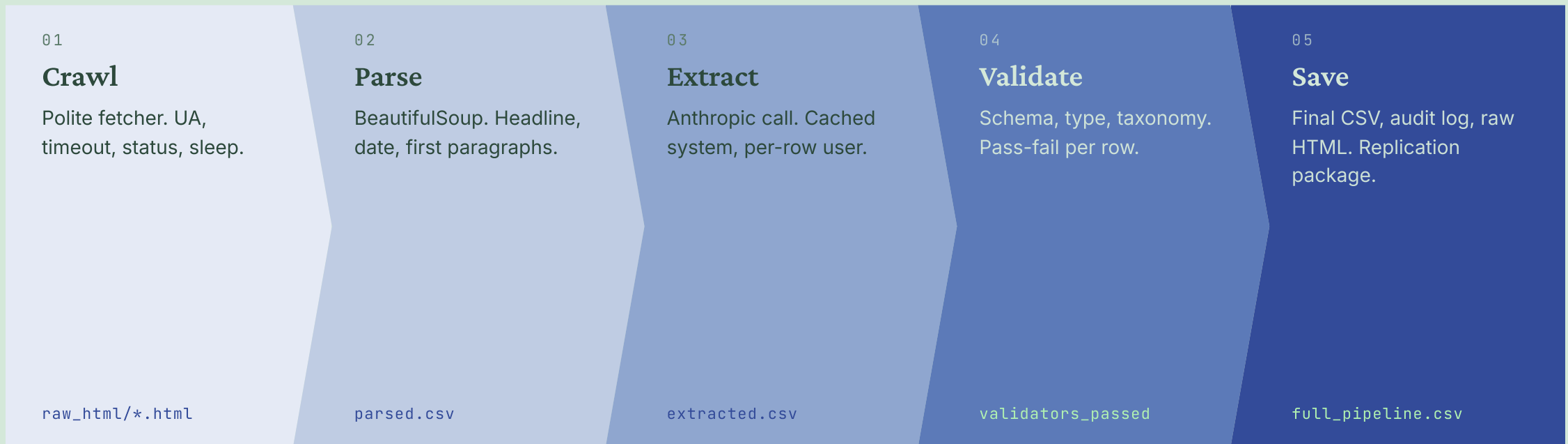
*Crawl, parse, extract, validate, save. Run together on five fresh articles.*

04

FIVE STEPS, EACH ONE WRITES TO DISK

# Architecture decision. Every step is restartable.

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*Each step reads from disk, writes to disk. If extract fails, you do not re-crawl. You re-extract.*

# Five fresh CSIS articles through all five steps.

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## What we run

Section 11. Five article URLs from the analysis index. Crawl with one-second sleeps. Save raw HTML. Print the manifest.

Section 12. Parse each raw HTML with BeautifulSoup. Pull headline, date, first three paragraphs. Save the cleaned CSV.

Section 13. Run the extractor on each parsed row. Cached system prompt. Per-row user message. Save extraction results next to the parsed text.

Section 14. Run validators. Print the pass rate. Print one full row from URL through validators-passed for inspection.

If any URL has changed since the workshop was prepared, switch to cached fallback HTML in the workshop GitHub repo at `day3/cache/`.

# What gets saved. Twelve columns. One row.

---

```
# One full row, all columns
url: https://www.csis.org/...
fetched_at: 2026-04-28T09:14:22Z
status: 200
raw_path: raw_html/ai-and-future...html
headline: "AI and the Future of Mediation"
date_text: Apr 27, 2026
n_paragraphs: 14
extracted_json: {"signatories": ["UN", "WTO"],
  "agreement_type": "technology",
  "year": 2026,
  "sector": "technology"}
model_id: "claude-haiku-4-5-20251001"
system_prompt_hash: a1b2c3d4...
input_tokens: 412
output_tokens: 48
validators_passed: True
```

## WHY THIS ROW

A reviewer can take this row and trace it back to a fetched HTML file, to a parser version, to a model version, to an exact prompt, to a token cost.

## TWELVE MONTHS FROM NOW

You will thank the past version of you who saved every column. The cost of saving more is zero. The cost of saving less is a request for the appendix you did not write.

Provenance for one row equals provenance for the dataset. The pipeline either saves it for every row or it saves nothing usable.

# Which model tier do you actually need?

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CLAUDE • CHEAPEST, FASTEST

## Haiku

Quick and cheap, built for high-volume, well-scoped jobs. The tier we run today.

**Use for** bulk extraction, normalization, tagging.

CLAUDE • BALANCED

## Sonnet

Stronger reasoning at a middle price. Reach for it when Haiku slips on harder calls.

**Use for** nuanced calls, longer context.

CLAUDE • MOST CAPABLE, PRICIEST

## Opus

The most capable and the slowest. Save it for the hardest reasoning or for building a gold standard.

**Use for** hardest cases, adjudication, gold.

**The rule.** Start at the cheapest tier, check it against your validators or a small gold set, and climb only if accuracy is short. Most wrangling never leaves the bottom rung.

Every provider offers the same ladder. OpenAI pairs lightweight tiers like GPT-5 mini and nano with a flagship like GPT-5 and reasoning models like o3 and o4-mini. The names change. The shape does not.

# Batch processing, for when you are not in a hurry.

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## WHAT IT IS

A normal call is synchronous. You send one request and wait a moment for the answer, which is what our demo does fifty times in a loop.

A batch call hands the provider a whole file of requests at once. It works through them in the background, and you collect the results later, usually within a day.

## WHY IT IS CHEAPER

Anthropic and OpenAI both bill batch work at roughly half the per-token price, for no change to your prompt or schema.

## REACH FOR IT WHEN

- You have thousands or millions of documents to score in one pass.
- An overnight turnaround is fine.
- Your prompt and schema are already settled.

## SKIP IT WHEN

- The work is interactive or time-sensitive.
- A synchronous run would finish in minutes anyway.

**The workflow.** Validate on a handful of records synchronously, the way we did today, then send the full corpus as one batch and halve the bill.

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CLOSING • 20 MINUTES

# Reproducibility.

*The appendix your reviewer is looking for. The arc you just completed. What to do this week.*

05

# The reproducibility appendix for an LLM-based measurement paper.

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## MUST-HAVE

- The full prompt verbatim, system and user, with date finalized.
- Model ID with version pin. Today's pin is `claude-haiku-4-5-20251001`.
- Temperature and any other generation parameters set.
- Schema with named taxonomies.
- Schema-validation pass rate on the full corpus.
- Robots.txt at date of crawl, plus your User-Agent string.
- Raw-HTML files in the replication package, or a stable archive mirror.
- Audit log with one row per call.

## OFTEN ASKED FOR

- A small hand-coded gold set with field-level F1 against the model.
- A determinism rate from running the pipeline twice.
- Subgroup pass-rate breakdown. Sector, region, length.
- Total cost summary at production scale.
- Simpler-baseline comparison. Rule-based or smaller model.

*If column one is in the appendix, the pipeline is defensible. Column two is where revisions go.*

# The three-day arc.

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## DAY 01 • THE MODEL ZOO

### Two families, one decision

You can name the encoder and generative families and articulate the choice. You ran a state-of-the-art encoder on real political text in your browser.

## DAY 02 • THE MEASUREMENT

### From notebook to proxy

A codebook, a thirty-paragraph gold set, a thirteen-hypothesis ensemble, a country-year proxy plotted against external events, a validation appendix.

## DAY 03 • THE PIPELINE

### Wrangling and crawling

A structured extraction pipeline, a polite crawler, a five-step end-to-end pipeline, an audit log that ships with the paper.

The encoder and the generative model do different jobs in the same project. The hard part of measurement work in 2026 is the codebook, the gold set, the validation, and the audit. Those are the same on both sides. The model just changes which knob you turn.

# Three days. Two families. One audit log.

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## THREE THINGS TO DO THIS WEEK

- Re-run the Day 3 notebook on five articles of your own choosing.
- Write a one-page replication appendix using slide 28 as the template.
- Pick the next concept. Spend an hour with a colleague writing the codebook.

## OPTIONAL READING

Grimmer, Roberts, and Stewart, *Text as Data* (Princeton 2022). The reference text for measurement work in our field. Validation chapters map directly onto the appendix you just learned to ship.

Replication package. [github.com/jacqpark/instats-llm-workshop](https://github.com/jacqpark/instats-llm-workshop)

*Thanks for spending three days here.*

**Stop**